Friction adjusts before sliding begins

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Static friction / ES: fricción estática / PT: atrito estático

Before sliding. Adjusts to match the pull, up to a limit.

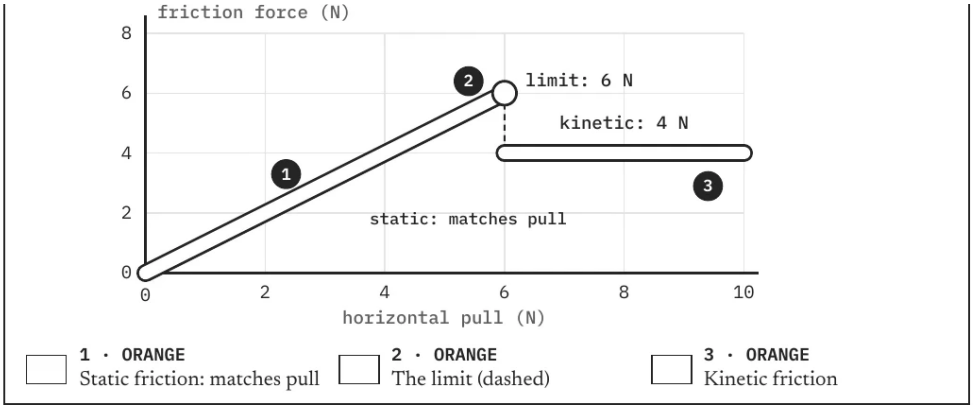
Also on your note page: Maximum static friction, Kinetic friction, Coefficient (μ).

B READ THE RELATIONSHIP

Static friction acts between surfaces that do not slide past each other.

Static friction adjusts up to a limit; sliding friction acts during slip

C TRACE THE ORIGINAL FIGURE



Use the numbered key. Trace one feature. Tell your partner what it means.

D FOLLOW THE WORKED EXAMPLE

A 2 kg crate slides right. Pull = 10 N right; kinetic friction = 4 N left. Find its acceleration.

Given m = 2 kg pull 10 N right f k 4 N left

Rule a = F net ÷ m

Work F net = 10 − 4 = 6 N right a = 6 ÷ 2

Answer 3 m/s² right

E SAVE FOR THE FINAL MEETING

Friction needs surfaces rubbing together. In space there are none to slow XJ down, one more reason it keeps its speed between observations.

Read the physics: Static friction

SUPPORTED READING / read one part, then point to the evidence Name

01 PICTURE IT

Push gently on a crate. If it stays still, static friction can balance the pull. A harder push needs more static friction, up to a maximum.

02 HOW IT WORKS

After slipping begins, use kinetic friction. In this model its size is the kinetic coefficient times the normal force. It opposes relative slipping at the contact.

03 WHAT TO CHECK

On a force trace, find the peak just before slip. Compare it with the steady-speed pull while the surfaces slide. Do not use the maximum static force for every pull.

04 TRY THE EXAMPLE FROM YOUR NOTES

A 2 kg crate slides right. Pull = 10 N right; kinetic friction = 4 N left. Find its acceleration. 3 m/s² right.

05 CONNECT TO XJ

Friction needs surfaces rubbing together. In space there are none to slow XJ down, one more reason it keeps its speed between observations.

SAY IT BEFORE YOU WRITE

When do you switch from static to kinetic friction? Before sliding, friction ___. During sliding, we use ___.

Build the word. Draw the idea.

STUDENT CHOICE / FRONT / OPENING MEETING Name _____

A FRAYER MODEL

Static friction / ES: fricción estática / PT: atrito estático

Say it in any language. Draw or use a few words. Use the notes to check your idea.

Meaning	How can you tell?
To me, static friction means …	You push a crate sideways and it stays still. What is happening where the surfaces touch? The surfaces stay ___. Friction acts toward ___.
Example or sketch	**Non-example**
I see it when …	This looks close, but …

B DRAW WITH YOUR TEACHER

Sketch a force trace. Label the peak just before slip and the steady-speed pull while the surfaces slide.

Labels you can use: at rest · just before slip · sliding.



Tell a partner: Before sliding, friction ___. During sliding, we use ___.

On assignment: Static friction

SECTION E / BACK / FINAL MEETING ONLY Name _____

A show cuts from a block sliding on a rough table to XJ coasting in empty space. The proposed caption says "Both will stop from surface friction."

PICK A JOB / DELIVER ONE PRODUCT

☐ **Effects engineer**

Specify the friction setting for each scene. Mark the contact surfaces, then tell the animator which motion needs another force to change.

☐ **Science broadcaster**

Record a correction for the cut. Your producer points to the surfaces in each scene and asks what could still bend XJ's path.

Word bank: at rest · just before slip · sliding.

Try: "Our desk recommends ___. This mark shows ___."

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

"We first said ___. Now we ___ because ___."